

Weakly Supervised Semantic Segmentation of Mangrove Ecosystem Using Sentinel-1 SAR and Deep Convolutional Neural Networks

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Abstract—Mangrove ecosystems are vital blue carbon habitats, and their accurate mapping and ongoing monitoring are essential for conserving their unique ecological and environmental benefits. This study employed a weakly supervised deep learning framework for semantic segmentation of the ecosystem by integrating time-series Sentinel-1 Synthetic Aperture Radar imagery with an existing ecosystem map as weak labels. This approach incorporates a U-Net architecture with an EfficientNet-B7 backbone for robust feature extraction, employs Test-Time Augmentation (TTA) with an ensemble strategy to enhance prediction stability, and validates the method's temporal transferability across multiple years. The model achieved high accuracy in the baseline year (2020), with 95.56% overall accuracy according to independent reference samples, and successfully generated noise-free maps despite imperfect training data. The TTA approach improved the final map both visually and statistically compared to a simple prediction approach. Visual comparisons with high-resolution satellite imagery, an existing ecosystem map, and a recent global mangrove extent dataset confirmed the superior performance of our proposed workflow in mangrove mapping. The temporal transferability of the model, examined in 2017 and 2023 using the baseline model with no modification, confirmed its applicability by generating the ecosystem maps meticulously, though the predictive uncertainty value increased. The generated multitemporal maps indicated a mangrove extent expansion of about 3.43% between 2017 and 2023. This straightforward workflow facilitates automatic mangrove ecosystem mapping and updating while effectively reducing the need for frequent reference sample collection, especially in cloud-prone areas. The temporal capability of the approach offers practical solutions for large-scale mangrove ecosystem conservation initiatives.

Index Terms—Deep convolutional neural network (CNN), mangrove, semantic segmentation, Sentinel-1, synthetic aperture radar (SAR), U-Net, weak supervision.

I. INTRODUCTION

MANGROVE ecosystems, recognized as blue carbon ecosystems, have a remarkable capacity to sequester carbon and mitigate global warming. These ecosystems are predominantly located in tropical and subtropical regions at the interface of land and sea within brackish environments [1]. Mangroves and mudflats, as two land covers in these ecosystems, are great sources of carbon sinks with a two to four times higher capacity than terrestrial forests on average [2], [3], [4]. The diverse land covers within these ecosystems, including mangroves, mudflats, tidal zones, water bodies, and barren areas, provide suitable habitats for a range of aquatic and terrestrial fauna. Generating and updating accurate and detailed databases and maps of mangrove ecosystems is essential to preserve their ecological and environmental benefits. This is because different land covers within such ecosystems function synergistically with their neighboring habitats due to their trophic and hydrological connections. Consequently, considering all land cover types is necessary to support sustainable planning for ecological balance and ecosystem service preservation [5], [6].

Remote sensing has garnered significant attention from government agents and researchers working in mangrove ecosystem-related fields [7], [8], [9], [10]. It provides cost-effective synoptic views over large mangrove ecosystems with harsh environments and has efficiently contributed to understanding the ecosystems' distribution, extent, health, and dynamics over time [11]. Remote sensing sensors, including multispectral, hyperspectral, thermal infrared, light detection and ranging (LiDAR), and synthetic aperture radar (SAR), deliver data from different perspectives useful for mangrove ecosystem studies, such as species mapping [12], plantation allocation [13], and biomass estimation [14]. Among these, SAR sensors are advantageous in cloud-prone areas due to their capability to penetrate clouds and acquire consistent, all-weather data. The backscattered SAR signals are sensitive to vegetation structure and dielectric properties, factors that are highly relevant for distinguishing land cover types in coastal wetland and mangrove ecosystems [15]. For instance, the sensitivity of SAR signals to vegetation structure allows for accurate discrimination of

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mangroves from other coastal vegetation due to their unique canopy density and vertical complexity. Likewise, the sensitivity to dielectric constant enables differentiation of saturated mudflats and tidal zones from the drier barren areas surrounding the ecosystem. These physical interactions form the basis of SAR sensors' applicability in coastal wetland and mangrove ecosystem mapping. Recent advances (e.g., such as higher spatial resolution, multipolarization capabilities, multiple frequency bands, and a wide range of acquisition modes) have further enhanced the role of SAR in wetland mapping, both inland and coastal, with studies demonstrating its capability to support large-scale, long-term monitoring of dynamic ecosystems [16], [17], [18].

The significant progress of machine learning algorithms over recent decades has created exceptional opportunities to automatically exploit remote sensing imagery [19], [20]. In particular, the conjunction of remote sensing imagery, machine learning, and reference samples, either collected through field campaigns or visual interpretations, supports the successful execution of desired tasks, such as mangrove ecosystem mapping. For instance, Bihanta Toosi et al. [21] combined single-date multispectral images of Sentinel-2 and Worldview-2 with field-collected reference samples and a random forest (RF) classifier to generate a mangrove ecosystem map in eight classes, and they obtained an accuracy of 65.5%. Likewise, Ashiagbor et al. [22] incorporated Sentinel-1 SAR imagery for mangrove mapping in the Keta Lagoon Complex using a Support Vector Machine (SVM) classifier and in-situ collected field samples, achieving an overall accuracy of 89.28%. Most previous studies emphasized the applicability of time series imagery for accurate mangrove mapping, primarily due to their dynamic nature, driven by tidal fluctuations [23]. In such dynamic environments, time series imagery allows machine learning algorithms to learn patterns in different tidal situations, enhancing classification accuracy.

Despite the prominent performance of machine learning algorithms, several studies reported the lower applicability of such algorithms and the necessity of developing more sophisticated ones. For instance, Ghorbanian et al. [24] demonstrated the weak performance of combining time series SAR imagery and RF for mangrove ecosystem mapping in nine classes, as the produced map obtained a relatively low accuracy of about 62.68%. A similar conclusion was reached by Aja et al. [25], in which the utility of RF and SAR images reached an accuracy of 84.6% in a four-class framework, while multispectral images resulted in an accuracy of 98.9%. In the meantime, deep learning models attracted considerable attention due to their high efficiency in handling complex remote sensing tasks [26], [27], [28]. Accordingly, researchers developed new methodologies shaped by deep learning models for mangrove studies. de Souza Moreno et al. [29] utilized dense (i.e., all scenes within a year) time series of Sentinel-1 images to delineate mangrove extent in parts of Brazilian coasts in São Paulo, where U-Net and EfficientNet performed promisingly for mangrove delineation with 97.35% accuracy. In another study, the authors employed time series multispectral images of Sentinel-2 for mangrove extent dynamic mapping in two restoration initiatives [30]. Their results revealed that U-Net reached an accuracy of 92.72%, outperforming SVM.

Likewise, a few studies were conducted to generate detailed mangrove ecosystem maps using deep learning models, almost all of which focused on using multispectral images [31], [32], [33]. Accordingly, further investigations are necessary to evaluate the performance of combining deep learning models and SAR imagery for detailed mangrove ecosystem mapping. The intention of using SAR sensors is associated with their frequent image acquisition without atmospheric and cloud interferences, enabling consistent mapping and monitoring. However, the requirement of numerous reference samples limits the applicability of deep learning models for supervised semantic segmentation tasks. One possible solution is to adopt a weak supervision framework, which eliminates the need for highly accurate reference samples by employing less precise labeled samples (i.e., weak labels). Such a framework leverages partial or imperfect labels to perform supervised learning [34], [35]. This not only reduces the burden of collecting accurate reference samples but also provides the numerous samples required for deep learning models. In such a context, existing land cover maps with suitable accuracy, with or without refinements, can be used as image-level weak labels to support the supervised classification task [36], [37]. Although weak supervision frameworks have been widely used for general land cover classification, few studies have explored their application to mangrove or wetland ecosystems [38], [39]. For instance, Otsu and Zhou [39] proposed a weakly supervised deep learning framework for mangrove community segmentation using noisy labels and WorldView-3 imagery. Their approach employed a ResNet50 backbone within a DeepLabV3 architecture, demonstrating the potential of weak supervision.

Regular mapping and updating of mangrove ecosystem status are essential to maintaining practical conservation efforts. Giri et al. [40] analyzed mangrove extent dynamics in the United States, harnessing the long-term archive of Landsat imagery between 1980 and 2020. They devoted considerable time and effort to manually collecting reference samples through visual interpretation of images for each time period, supporting a supervised classification scheme. Similarly, Zhang et al. [41] collected reference samples through field campaigns and visual interpretations on an annual basis to monitor the spatial and temporal change of mangroves in Mangrove National Nature Reserves, China, using Sentinel-2 imagery. A key limitation of such approaches is the necessity of the existence of reference samples for each studied period, the collection of which is laborious and requires considerable time and effort. Other solutions include migrating reference samples from a baseline year to a target year through change analysis or migration equations [42], [43], [44], which proved to be successful, though they require additional computations and human intervention for tuning. Another alternative solution is to rely on the learning potential and transfer learning capability, e.g., in time, of deep learning models, which can contribute toward eliminating the necessity of multiannual reference sample collection for time series analysis [45], [46].

Considering the preceding, this study advances mangrove ecosystem mapping through two main objectives. First, it investigates deep learning based semantic segmentation of mangrove

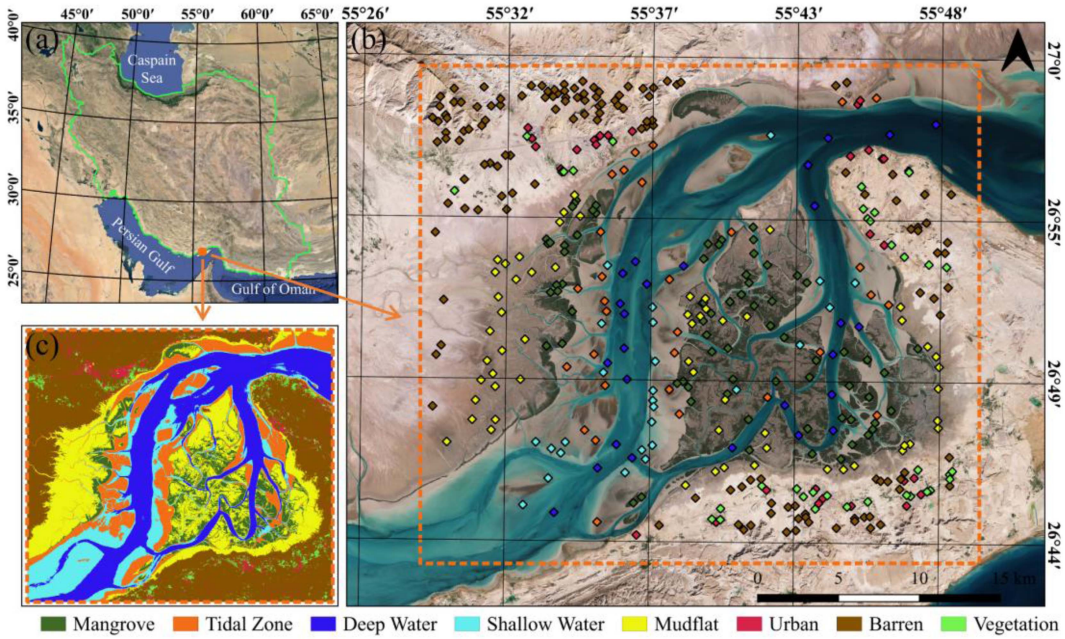


Fig. 1. (a) Geographical location of the study area in Iran and the Persian Gulf. (b) Spatial distribution of the test reference samples. (c) Pre-existing ecosystem map of the study area 9.

ecosystems using Sentinel-1 SAR imagery in a weakly supervised fashion, addressing prior limitations of conventional machine learning algorithms and SAR for this task [24], [25]. Additionally, few studies have explored weakly supervised learning for mangrove ecosystem classification, underscoring the need for further research. Second, the study applies the trained model from a baseline year to past and future data for automated mangrove ecosystem mapping, assessing the temporal transferability of the model. This approach enables continuous monitoring of ecosystem changes while eliminating the need for collecting multiannual reference samples. Here, the focus is on detailed mangrove ecosystem mapping by considering various existing land cover types rather than only delineating mangrove extents, a focus that has been neglected in most previous studies. To this end, time series Sentinel-1 images, along with an existing land cover map as weak labels, are used to train a deep learning model in a weakly supervised fashion to map the mangrove ecosystem in Qeshm, Iran, in the baseline year of 2020. After statistical assessments using independent reference samples collected through visual interpretation of high-resolution satellite images, the temporal transferability of the model for mangrove ecosystem mapping is evaluated by generating the ecosystem maps for 2017 and 2023.

II. MATERIALS AND METHODS

A. Study Area

The study area extends from $26^{\circ} 42' N$ to $26^{\circ} 59' N$ longitudes and $55^{\circ} 28' E$ to $55^{\circ} 49' E$ latitudes, located within the northwest estuaries of Qeshm Island, south of Iran (Fig. 1). It encompasses the largest mangrove ecosystem in the Persian Gulf [47], the Hara Biosphere Reserve, designated as a protected area under

the Ramsar Convention [48]. The area is recognized for its rich biodiversity, and being a critical habitat for various aquatic and terrestrial fauna, serving as a sanctuary for migratory birds. The Hara Biosphere Reserve dominantly accommodates *Avicennia marina*, though the existence of *Rhizophora mucronata* has been reported in the Persian Gulf mangrove communities [49]. The tidal cycle of the study area is semidiurnal, with considerable fluctuations ranging from 0.3 to 4.6 m [50], resulting in the generation of many tidal creeks and vast intertidal zones devoid of surface vegetation.

B. Datasets

1) *Reference Datasets*: As required by this study, two datasets were used to support the training, validation, and testing processes of the proposed methodology.

The first dataset was a pre-existing ecosystem map of the study area used as a weak label mask (i.e., an image determining the class of each pixel) for the deep learning model development. This map was produced according to the suggested workflow by Ghorbanian et al. [24], with the potential to achieve high classification accuracy over mangrove ecosystems. According to the method, seasonal Sentinel-1 and Sentinel-2 images were inserted into an RF algorithm, along with pre-existing reference samples of eight classes. Although the ecosystem map contained some errors and uncertainties associated with pixel-based classification, it is anticipated that deep learning models can alleviate the impact of noisy labels and work well with imperfect label masks [51].

Following this, the second dataset, called test reference samples, included polygon-based samples used to independently evaluate the mangrove ecosystem map generated by the deep

learning model, the centers of which are presented in Fig. 1(b). These samples were manually collected for the baseline year of 2020 through visual interpretation of true-color high-resolution satellite images. In total, 415 polygons with an area of approximately 160.2 ha containing eight classes of mangrove (24.8 ha), tidal zone (18.8 ha), deep water (20.2 ha), shallow water (15.7 ha), mudflat (21.1 ha), urban (19.9 ha), barren (24.6 ha), and vegetation (15.1 ha) were collected. Homogeneous locations were considered for the initial sampling to avoid mixed pixel challenges in fragmented locations, followed by adding several mangrove samples from narrow patches and sparse locations. This procedure allows for a better assessment of the produced map regarding the mangrove delineation. This is rooted in the fact that some mangroves are located along the tidal zones or grow nonuniformly. Notably, the test reference samples were spatially disjointed from the ones employed to generate the pre-existing ecosystem map [i.e., weak label mask in Fig. 1(c)].

2) *Satellite Dataset*: Sentinel-1 carries a CSAR instrument (5.405 GHz), allowing for all-weather data acquisition in four modes [52]. Sentinel-1 is a joint development between the European Commission Initiative Copernicus and the European Space Agency to support the Global Monitoring for Environment and Security program. Time series of Sentinel-1A Ground Range Detected (GRD) images from the Interferometric Wide Swath mode captured in both ascending and descending orbits were used. Each GRD image comprises two polarizations of VV (vertical transmittance and vertical receiving) and VH (vertical transmittance and horizontal receiving) at a 10-m spatial resolution. All available Sentinel-1A GRD images over the study area between 1 January 2020 and 1 January 2021, comprising 22 (March–May), 23 (June–August), 23 (September–November), and 21 (December–February) images in Spring, Summer, Autumn, and Winter, respectively, were employed for mangrove ecosystem mapping. In dynamic environments like mangrove ecosystems, where tidal fluctuations occur, time series imagery enhances the delineation of existing classes [23]. The intention of using SAR sensors is associated with their frequent image acquisition without atmosphere and cloud interferences, enabling consistent mapping and monitoring, especially in cloud-prone regions. The Sentinel-1A GRD images were acquired from the Google Earth Engine (GEE) cloud computing platform [53]. Similarly, the satellite datasets for the years 2017 and 2023 were generated to allow for temporal transferability analysis of the trained deep learning model.

C. Methodology

Fig. 2 presents the proposed workflow for mangrove ecosystem mapping using a time series Sentinel-1 dataset and a deep learning model. This workflow outlines the systematic approach of the task, with details of each step provided in separate sub-sections.

1) *Satellite Dataset Preprocessing*: Sentinel-1A GRD images contained in GEE are accessible using the snippet of `ee.ImageCollection("COPERNICUS/S1_GRD")`. Five preprocessing steps of (1) updating orbit file metadata, (2) removing low-intensity noise on scene borders, (3) removing additive

thermal noise, (4) computing backscattering values, and (5) correcting the terrain effect using Shuttle Radar Topography Mission Digital Elevation Data data have been initially applied by GEE developers. These common preprocessing steps make the Sentinel-1A GRD images ready to use for many applications. Finally, a mean function was applied to all images within each season to generate seasonal SAR features containing VV and VH polarizations for each season. The mean function was applied as it best represents the general coastal zone conditions by reducing speckle noise and transient anomalies while preserving characteristic seasonal backscatter patterns of mangrove ecosystems [15], [24]. Accordingly, the final satellite dataset in the baseline year (i.e., January 2020 to January 2021) included eight SAR features, which were used for mangrove ecosystem mapping.

2) *Deep Convolutional Neural Network Model*: U-Net is a fully convolutional neural network with a proven capability of handling remote sensing images for semantic segmentation tasks [54], [55]. The U-Net has a symmetrical U-shape architecture comprising an encoder, a decoder, and skip connections [56]. The encoder applies down-sampling filters, reducing the spatial resolution and increasing the spectral resolution, while the decoder pass does the exact opposite. Subsequently, high-level semantic features (in the encoder) and low-level features with high spatial information (in the decoder) are concatenated through skip connections. Integrating low-level and high-level features at different scales empowers the model to efficiently learn the patterns at multiple scales, enabling better identification of land covers and improving the semantic segmentation results [57], [58]. Here, the EfficientNet-B7 model was used as the backbone (i.e., encoder and decoder) of the U-Net architecture due to its efficiency in feature extraction. EfficientNet-B7 is a variant of the EfficientNet family, with 33 layers and the MBConv blocks as the fundamental component, which is reinforced with Squeeze and Extraction blocks for efficient feature extraction [59], [60], [61]. This family benefits from the compound scaling method, in which three parameters of network depth, width, and resolution are balanced simultaneously [60]. The compound scaling enables the model to capture intricate patterns in various land covers, such as mangrove boundaries, water bodies, and urban areas. For simplicity, the combination of U-Net and EfficientNet-B7 will be referred to as UNB7 from now on.

3) *Mangrove Ecosystem Semantic Segmentation (Implementation Details)*: In the first step, the seasonal SAR features (satellite dataset) were individually normalized by subtracting the mean and dividing by the standard deviation, setting the mean to zero and values to be bound between -1 and 1 . This step is critical for optimizing training efficiency and improving the generalization capability of deep learning models [62]. Subsequently, the SAR features and the weak labels were, respectively, cropped into patches of $64 \times 64 \times 8$ and $64 \times 64 \times 1$ without any overlap between neighboring patches, resulting in 2592 patches.

Fig. 3 shows sample patches (SAR features and corresponding weak labels) of the study area, along with their true-color high-resolution satellite images for better interpretation. Later, the weak labels were encoded using the one-hot encoding scheme to generate patches of $64 \times 64 \times 8$ (where eight corresponds to the number of classes) for further processing. The patches were

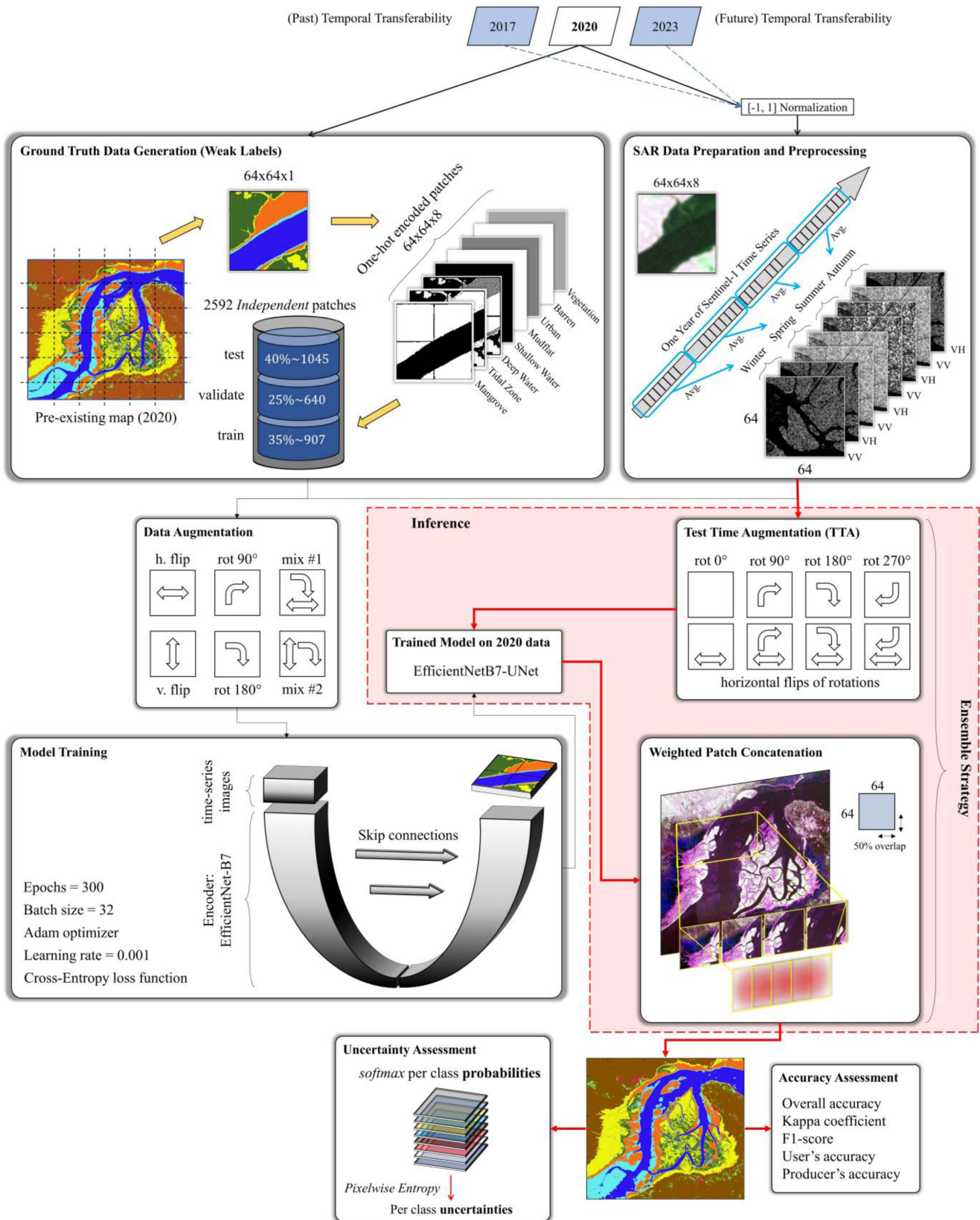


Fig. 2. Schematic overview of the developed workflow for mangrove ecosystem mapping using time series Sentinel-1 and a deep learning model.

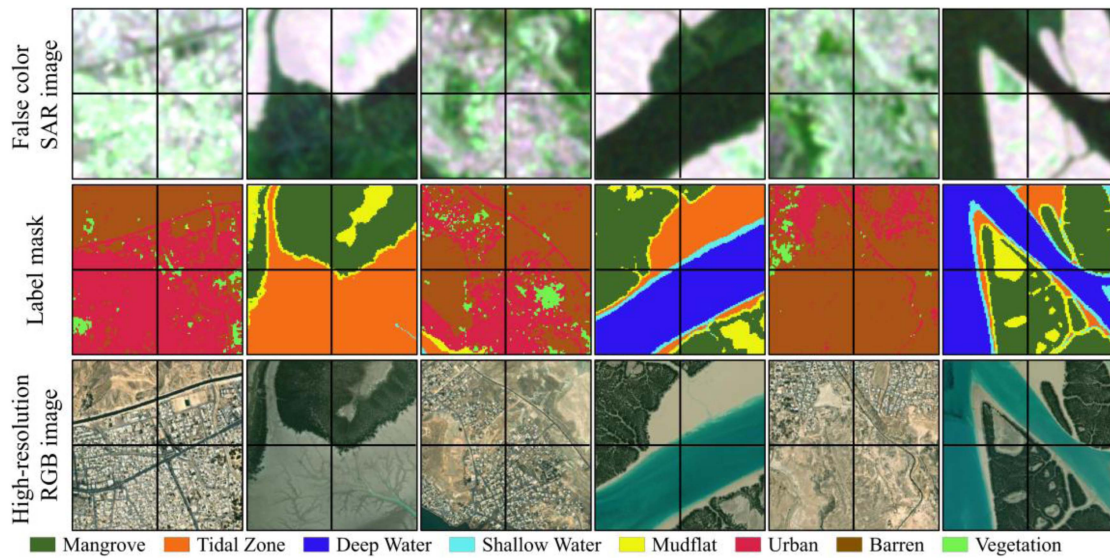


Fig. 3. Sample patches across the study area presenting false color SAR images (first row), label masks (second row), and the corresponding high-resolution RGB images (third row) for better visualization of the considered classes for mangrove ecosystem mapping.

then randomly divided into three sets of training, validation, and test patches with ratios of 35% (907 patches), 25% (640 patches), and 40% (1045 patches), respectively. Data augmentation was applied to expand the training patches to enhance the performance of the model and reduce the risk of overfitting. In this regard, two data augmentation techniques of rotation and flip, along with their combinations, were applied. Finally, the augmented training set was used to train the UNB7 model (see Section II-C-2), with random initial weights, a batch size of 32, cross-entropy as the loss function, and the Adam optimizer set with a learning rate of 0.001. The model was trained in 300 epochs, and the best model, identified based on the overall accuracy of the validation set, was saved for further processing.

Afterward, the mangrove ecosystem map for the entire study area was generated by applying the best model to all patches. In this regard, an ensemble strategy was adopted to reconstruct the final mangrove ecosystem map instead of simply concatenating classified patches within a Test-Time Augmentation (TTA) framework [63], [64], [65]. Within this framework, the entire satellite dataset is augmented during the prediction phase, and predictions from all augmented versions are merged, improving model performance and robustness.

In the first step, the entire satellite dataset was padded horizontally and vertically with a 32-pixel margin (i.e., half of the patch size). Later, the padded satellite dataset was subjected to four rotations (0° , 90° , 180° , and 270°) and a flip of each rotation to generate an eight-fold dataset, enabling the model to perform semantic segmentation with different viewing angles (see Fig. 4). The eight-fold satellite dataset was cropped into patches of $64 \times 64 \times 8$ with an overlap of 32 pixels (i.e., 50% overlap), and then the patches were fed to the best UNB7 model for prediction. Then, the predicted patches were smoothly concatenated by applying a weighted 2D second-order spline function concerning the 50% overlap between adjacent patches (see Fig. 5). In other words, the central predictions of each patch were used with

higher weights to reconstruct an eight-fold mangrove ecosystem map. Finally, the 8-fold predictions were aggregated to generate the final mangrove ecosystem map covering the entire study area. This strategy can eliminate horizontal and vertical artifacts and reduce misclassification errors associated with the lower prediction capability of the model because of fewer pixels at the border of patches. The final mangrove ecosystem map was visually compared with regular patch-based concatenation to ensure the advantages of the ensemble strategy.

Following the second objective, the best UNB7 model, trained using the satellite dataset and corresponding weak labels in 2020 as the baseline year, was used to produce mangrove ecosystem maps for the past and future years. To achieve this, satellite datasets for 2017 (past) and 2023 (future) were prepared, similar to the satellite dataset in 2020 (see Sections II-B-2 and II-C-1). Afterward, the model was applied to these satellite datasets, and the ensemble strategy was used to generate the corresponding mangrove ecosystem maps in 2017 and 2023. This approach enables the investigation of the UNB7 and seasonal SAR features' temporal transferability, potentially eliminating the necessity of collecting reference samples for the mentioned years [46].

4) *Accuracy Assessment*: Accuracy assessment is crucial to ensure the quality of generated maps using quantitative measures [66]. To this end, first, the model was evaluated using the test patches, which were not seen by the model during the training phase, by calculating the overall accuracy. Second, the independent test reference samples were used to assess the accuracy of the final mangrove ecosystem map by calculating the confusion matrix. Later, overall accuracy, kappa coefficient, F1-score, user's accuracy (UA), and producer's accuracy (PA) were derived from the confusion matrix for a robust assessment of the results. Visual comparisons between the final mangrove ecosystem map, high-resolution satellite images, and the recent global mangrove extent map were conducted to complement the statistical assessments.

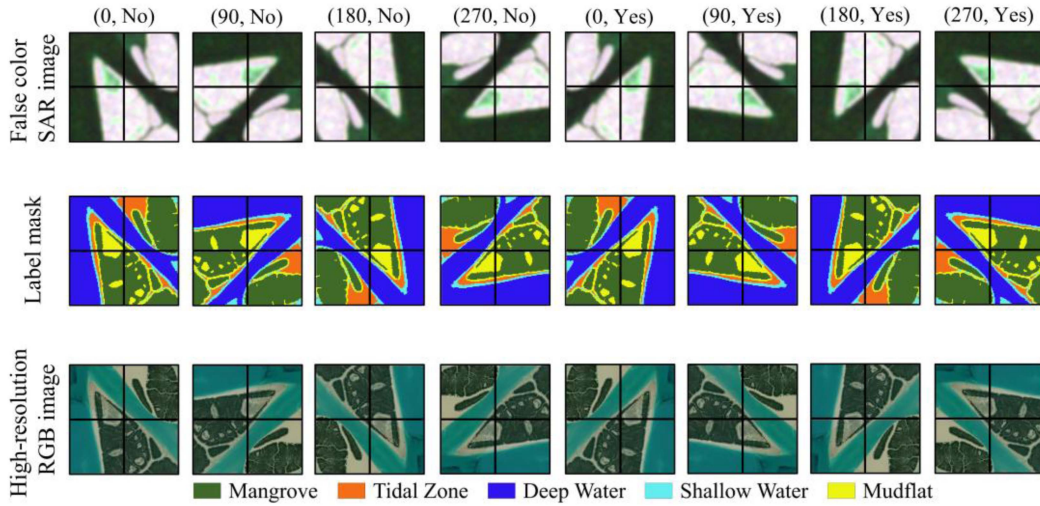


Fig. 4. Representation of the eight-fold satellite dataset generation within a TTA framework. The values in parentheses refer to rotation and horizontal flip augmentations, respectively.

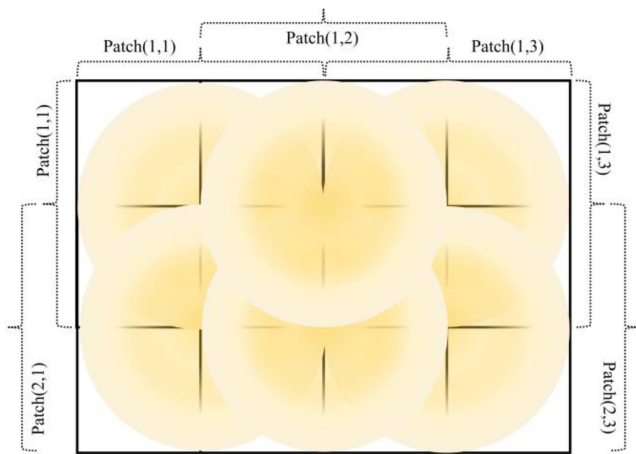


Fig. 5. Schematic representation of the weighted 2D second-order spline function used for concatenating adjacent patches with 50% overlap. Higher weights are related to darker yellow at the center of each circle, and lower weights are shown in lighter yellow.

III. RESULTS

A. Mangrove Ecosystem Mapping

As mentioned earlier, the model was trained for 300 epochs, and the best UNB7 model, based on the accuracy achieved over the validation patches, was selected for mangrove ecosystem mapping. The best UNB7 model, with an accuracy of 95.71% on the training patches, obtained an accuracy of 87.75% on the validation patches. It was then evaluated using the test patches, which were not seen by the model during the training phase, and an accuracy of 88.61% was achieved.

Afterward, the best UNB7 model was employed to generate the mangrove ecosystem map of the study area. In this regard, the ensemble strategy was adopted to reconstruct the final mangrove ecosystem map from the eight-fold classified patches [Fig. 6(a)].

Visually, the model was successful in generating an accurate map, especially focusing on the mangrove ecosystem (i.e., the central part of the study area), using the seasonal SAR images. In general, different classes were depicted meticulously, and the map provides an acceptable overview of the pattern of classes across the study area. For instance, mangroves were delineated precisely, and as anticipated, their surrounding pixels were assigned to the mudflat and tidal zone classes. The central parts of tidal creeks were classified as deep water, and water pixels in proximity to land areas (e.g., tidal zone, mudflat, and barren) were correctly assigned to the shallow water class. More specifically, as presented in Fig. 6(b), the proposed method was capable of providing an accurate map of the mangrove ecosystem when compared to a high-resolution satellite RGB image. It is evident that the model successfully distinguished different classes of mangrove, tidal zone, deep water, shallow water, and mudflat. Dense and sparse mangrove patches with open and closed canopies, as well as narrow mangrove patches at the edges of land areas, were delineated precisely, with higher reliability over dense mangrove patches. Moreover, the urban areas were identified with satisfactory accuracy, though occasionally, built-up areas were incorrectly classified as barren, as presented in Fig. 6(c). This misclassification happened mainly in urban areas with open low-rise building patterns. Furthermore, since the urban class covered a minor area, a relatively lower number of training patches with urban labels (with lower diversity in built-up patterns) existed to support the training phase of the model, and such an issue can be considered an influential factor for misclassifications. As illustrated in Fig. 6(d), the model had a relatively poor performance in regions with very sparse mangroves, mainly in cases of single scattered trees over mudflats. Similarly, it was observed that scattered terrestrial vegetation covers (i.e., outside the mangrove ecosystem) were incorrectly classified as barren, though most of the vegetation covers were correctly identified. This occurred over vegetation areas comprised of scattered trees and bushes, as found in arid

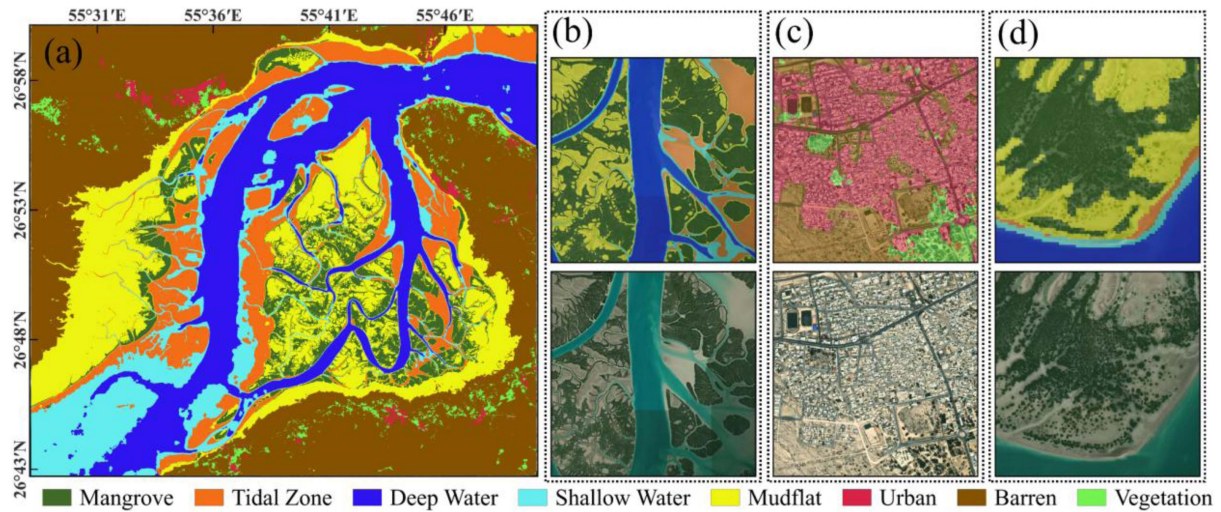


Fig. 6. (a) Mangrove ecosystem map of the study area reconstructed based on the TTA ensemble strategy and the deep learning model, and (b)–(d) zoomed results with corresponding high-resolution RGB images for better representation.

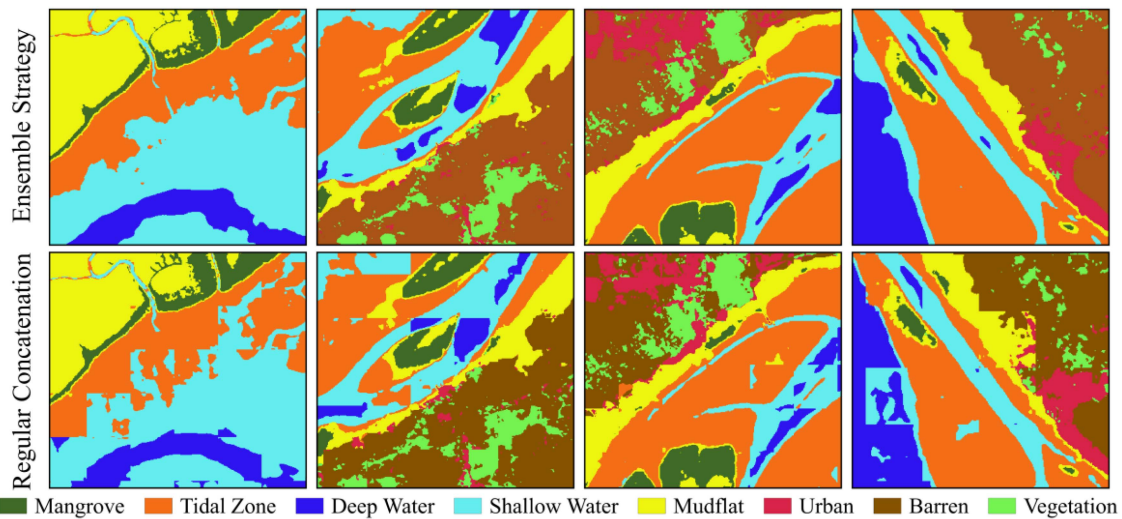


Fig. 7. Sample zoomed areas for visual comparison of the TTA ensemble strategy and regular concatenation for mangrove ecosystem map reconstruction.

and semiarid areas, where the high background effect of barren within the pixel size (10 m) of the satellite dataset impacted the classification.

Additionally, the final mangrove ecosystem map was visually compared to the pre-existing ecosystem map [Fig. 1(c)], which was used as the weak labels to train the deep learning model. As an output of a pixel-based classification without manual refinements, the weak label mask contained several instances of misclassification inside and outside the mangrove ecosystem. For instance, several pixels that belonged to the mudflat class were incorrectly classified as urban, scattered across the mangrove ecosystem. Likewise, salt-and-pepper misclassification errors existed between barren and urban classes. However, after a visual assessment, it was revealed that the model was capable of alleviating the noise label effects and generating a smooth mangrove ecosystem map. This confirmed the applicability of

the developed weakly supervised semantic segmentation workflow for mangrove ecosystem mapping since the workflow was successful in producing promising results even when nonideal weak labels were incorporated.

As mentioned earlier, the ensemble strategy was adopted to reconstruct the final mangrove ecosystem map. To investigate its advantage, a visual comparison was made between the map reconstructed based on the adopted strategy and the one shaped by regular patch concatenation. In this regard, the mangrove ecosystem map of the study area was also reconstructed according to the regular vertical and horizontal concatenation of the predicted patches ($64 \times 64 \times 8$) with no overlap. Four zoomed areas were then extracted from both maps to illustrate the differences between the two strategies (see Fig. 7). Overall, using the ensemble strategy, the results were more visually appealing compared to the regular concatenation, which created

TABLE I
CONFUSION MATRIX OF THE MANGROVE ECOSYSTEM MAP RECONSTRUCTED THROUGH THE TTA ENSEMBLE STRATEGY OF 8-FOLD SEASONAL SAR SATELLITE DATA AND THE DEEP LEARNING MODEL

	Mangrove	Tidal Zone	Deep Water	Shallow Water	Mudflat	Urban	Barren	Vegetation
Mangrove	2327	0	0	0	170	0	0	0
Tidal Zone	0	1828	0	36	0	0	0	0
Deep Water	0	0	2004	22	0	0	0	0
Shallow Water	0	120	55	1390	0	0	0	0
Mudflat	0	0	0	0	2121	0	0	0
Urban	0	0	0	0	34	1715	228	16
Barren	0	0	0	0	0	2	2434	0
Vegetation	0	0	0	0	0	15	14	1492
PA (%)	93.19	98.07	98.91	88.81	100	86.05	99.91	98.09
UA (%)	100	93.84	97.32	95.99	91.22	99.01	90.95	98.94
Overall Accuracy (OA) = 95.56%			Kappa Coefficient (KC) = 0.94			F1-Score = 0.95		

numerous horizontal and vertical artifacts. The existence of such artifacts may be associated with lower predictability of the model due to a lower number of pixels and a lack of enough contextual information at the borders of each patch [67]. The ensemble strategy fully resolved the issue by considering a 50% overlap between patches and using an 8-fold weighted approach for reconstruction (see Section II-C-3). As evident in Fig. 7, several horizontal and vertical discontinuities occurred at the border of patches during the concatenation, especially at the interface of land cover classes, such as tidal zone, deep water, and shallow water, or between mudflat, urban, and barren, though the mangrove class was minimally affected. Additionally, the lower predictability of the model at the border of patches led to several misclassifications when implementing the regular concatenation without considering the overlap between prediction patches.

The accuracy of the mangrove ecosystem map was then quantified using independent test reference samples, complementing previous visual interpretations. In this regard, the test reference samples were overlaid on the map, and the confusion matrix was calculated, as summarized in Table I. According to Table I, the UNB7 model achieved an overall accuracy, kappa coefficient, and F1-score of 95.56%, 0.94, and 0.95, suggesting the applicability of combining SAR data and the deep learning model for detailed mangrove ecosystem mapping. Moreover, the class-wise PAs and UAs were derived, and the UNB7 model obtained an average PA and UA of 95.37% and 95.90%. As is clear, deep water, mudflat, and barren classes obtained the highest PAs, while the mangrove class achieved the ideal UA. A minor portion of the mangrove reference samples was wrongly assigned to the mudflat (6.81%) class. The misclassification between mangrove and mudflat classes mainly occurred in areas with very sparse and scattered mangroves [see Fig. 6(d)], with dominant mudflat cover within Sentinel-1 pixels. On the other hand, the urban class achieved the lowest PA (86.05%) due to a relatively high number of wrongly classified pixels as barren. As mentioned earlier, such misclassifications were mainly associated with a limited number of pixels in the training patches with urban labels due to the low coverage of urban areas across the study area. Moreover, mutual misclassifications were observed between the tidal zone and shallow water classes, which were particularly related to the tidal situation of the mangrove ecosystem.

Afterward, the results of the developed workflow were compared with those obtained from an RF classifier. The RF classifier

was identified as a prominent machine learning algorithm in mangrove ecosystem mapping studies [68], [69]. Accordingly, the satellite dataset containing eight SAR features was given to an RF classifier for mangrove ecosystem mapping. The pre-existing reference samples, which had been used for weak label mask creation (see Section II-B-1), were used to train the RF classifier, and the out-of-bag error was used for hyperparameter tuning. Finally, the RF-generated mangrove ecosystem map was evaluated using the test reference samples, and an overall accuracy and kappa coefficient of 77.35% and 0.74 were yielded. The results demonstrated the lower applicability of the RF classifier in comparison with the developed workflow, as the overall accuracy and kappa coefficient values declined by about 19.05% and 21.27%.

B. Predictive Uncertainty

In addition to the mangrove ecosystem map in discrete form (i.e., categorical labels), class-wise probabilistic maps, shown in Fig. 8, were also derived for further investigations and studying the spatial performance of the UNB7 model. Generally, the model had lower probability values at the intersection of different classes, which is anticipated due to the challenge posed by mixed pixels. As evident in Fig. 8(a), mangroves were mostly delineated with high probability values, and in particular, nearly 90% of mangrove pixels exhibited a class probability value of above 80% during the prediction step. Likewise, the model confidently assigned labels to the tidal zone, deep water, shallow water, mudflat, and barren classes, though a minority of pixels with low probabilities were observed at the transitions between these classes. For instance, the UNB7 model yielded lower class probabilities in the borders of shallow water and tidal zones, which has resulted in several wrongly classified pixels possibly linked to tidal fluctuations [Fig. 8(b) and (d)], as also reported in Table I. In contrast, the probabilistic maps of two classes of urban and vegetation showed a relatively higher number of pixels with low probability [Fig. 8(f) and (h)]. In particular, fewer pixels assigned to these classes had high probability values, highlighting the challenge faced by the UNB7 model in accurately classifying these areas. This difficulty likely arose from the mixed pixel effect, particularly at the intersection of these classes, and the relatively lower number of training patches containing these classes. The urban areas, in particular,

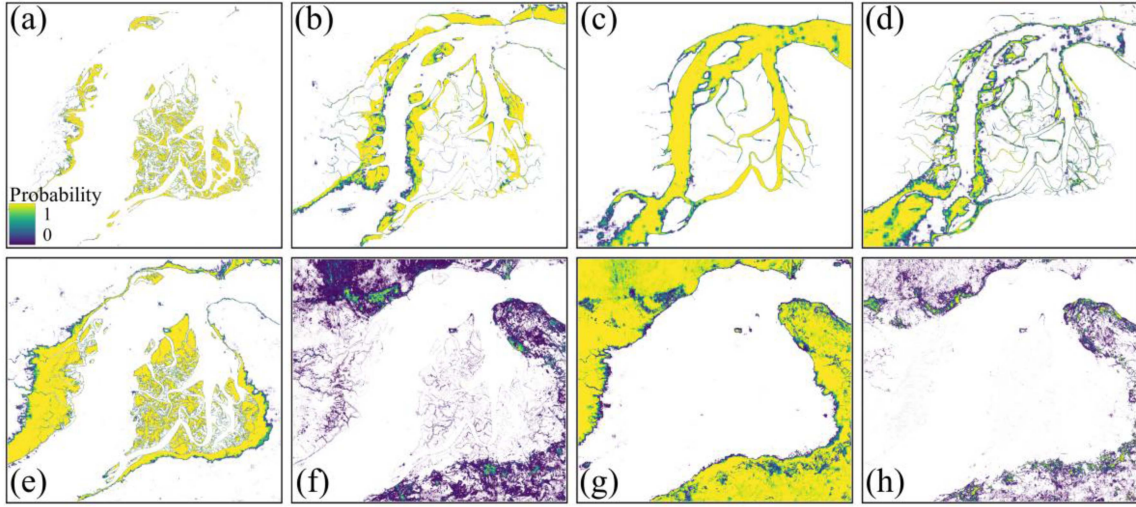


Fig. 8. Probability maps of the mangrove ecosystem based on the best UNB7 model, including classes of (a) mangrove, (b) tidal zone, (c) deep water, (d) shallow water, (e) mudflat, (f) urban, (g) barren, and (h) vegetation.

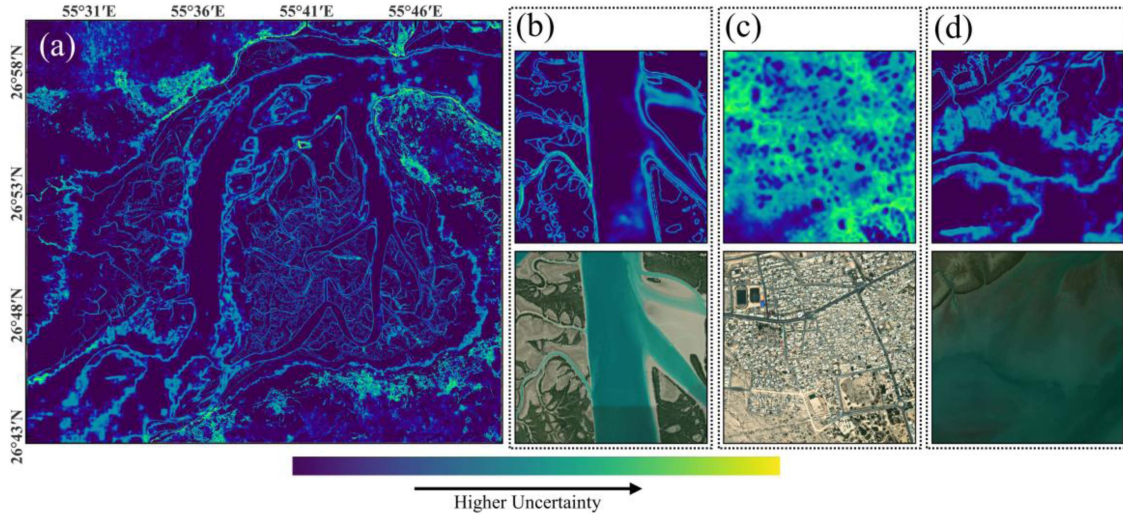


Fig. 9. (a) Uncertainty map of the study area calculated using the Shannon entropy and the probabilistic maps of the UNB7 model, (b)–(d) along with zoomed patches featuring corresponding high-resolution RGB images for better representation.

exhibited the lowest probabilities, reflecting the complexities of distinguishing urban areas from surrounding barren areas [see Table I and Fig. 8(f)], except in regions with compact building patterns. Another reason for such low probabilities could also be related to the weak label mask, which contained several wrongly classified pixels of barren lands as urban, especially in the northwestern parts of the study area with rough topography.

To further deepen the uncertainty analysis, the predictive entropy of the UNB7 model was calculated using the Shannon entropy ($H(p)$, (1) and the probabilistic maps [70], [71]. In (1), C is the total number of classes and p_i is the predicted probability of class i for a given pixel.

$$H(p) = - \sum_{i=1}^C p_i \log_2(p_i). \quad (1)$$

Fig. 9 shows the predictive entropy of the UNB7 model outputs, and it can be seen that higher uncertainty occurred at the borders of intersections between different classes. Moreover, minimal (close to zero) uncertainty was observed in central parts of the ecosystem, especially classes of deep water, mangroves, and mudflats. Relatively higher uncertainty values were noted in the northeastern and southwestern parts, where deep water and shallow water intersect. Higher uncertainty values were seen outside of the ecosystem, especially between urban and barren areas, where the entropy values reached nearly 1.6, indicating high uncertainty in the model's ability to assign labels to these classes. The mean entropy of the map [Fig. 9(a)] was about 0.1867, with pixels labeled as mangrove having the lowest share of entropy (5.62%) among all classes in the study area. On the other hand, the pixels that were assigned to barren had the

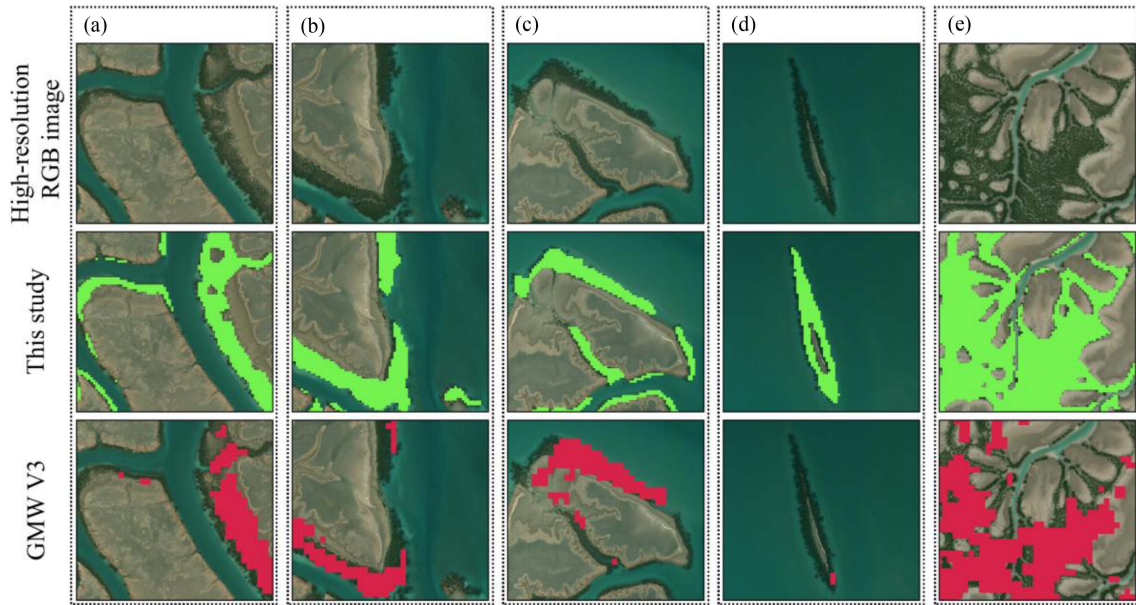


Fig. 10. (a)–(e) Visual comparison between the extracted mangrove extent map of this study and the Global Mangrove Watch Version 3 (GMW V3), overlaid on corresponding high-resolution RGB images over five sample regions in the mangrove ecosystem.

highest share of the total entropy (32.67%), as also evident in the provided Fig. 9(c). As evident in Fig. 9(b) and (d), the pixels with higher uncertainty in the ecosystem were related to the borders of shallow water and tidal zones, as well as shallow water and deep water, mostly related to the tidal fluctuations. The lower probability values in Fig. 8 also manifested themselves as high entropy values outside of the ecosystem, as shown in Fig. 9(c). The uncertainty analysis revealed better performance (higher confidence) of the UNB7 model within the ecosystem itself.

C. Comparison With Global Mangrove Map

A mangrove extent map was extracted from the multiclass mangrove ecosystem map generated by the UNB7 model for further investigation into the applicability of the developed workflow in mangrove delineation. This map was compared with the latest version of the global mangrove map in 2020. The third version of the global mangrove ecosystem map, known as Global Mangrove Watch (GMW V3), was considered for comparison [72]. The GMW map was generated using the Advanced Land Observing Satellite 2 (ALOS-2) Phased Array L-Band Synthetic Aperture Radar 2 (PALSAR-2) dataset with a spatial resolution of 25 m. Although the nature of datasets in the current study and the GMW are similar (both using SAR), their outputs differ significantly (see Fig. 10). Despite the difference in spatial resolution of the two maps, which might impact the results in very sparse and scattered mangroves, the results confirmed the superior performance of the UNB7, even in dense mangrove areas. As evident in Fig. 10, the proposed workflow outperformed the GMW in mangrove delineation on several occasions. The model performed more robustly when delineating not only dense mangrove patches but also the narrow patches at the edges of land areas, while the GMW was not

successful in delineating mangrove patches [Fig. 10(a)–(c)]. Furthermore, the mangroves around relatively small land areas were fully depicted by the model, while only a few pixels were identified as mangroves in GMW [Fig. 10(d)]. Likewise, the UNB7 model confirmed its capability in distinguishing the mangrove and mudflat classes since the mudflat holes between mangrove patches were identified precisely [Fig. 6(e)].

D. Temporal Transferability

The visual and statistical assessments in the previous sections confirmed the promising applicability of the UNB7 model for mangrove ecosystem mapping. Several analyses were comprehensively conducted in 2020, the baseline year, to ensure the robustness of the proposed workflow. However, the applicability of the trained UNB7 model for mapping the mangrove ecosystem in both earlier and future time periods was not assessed. In this section, the temporal transferability of the model was assessed by generating mangrove ecosystem maps in 2017 (past) and 2023 (future), focusing specifically on the mangrove ecosystem itself rather than on the entire study area. Regarding the temporal transferability analysis, the SAR datasets in 2017 and 2023 (see Section II-C-4), with three-year intervals from the baseline year, were fed to the trained UNB7 model in 2020. The generated ecosystem maps, as in Fig. 11, confirmed that the model indeed has notable temporal transferability potential. In particular, visual assessments and comparisons with available satellite images (i.e., Sentinel-2 and high-resolution images) in 2017 and 2023 demonstrated acceptable performance of the UNB7 model, though several misclassifications were observed. Additionally, for each mangrove ecosystem map of 2017 and 2023, 2000 stratified random samples were generated with appropriate spatial distribution across the ecosystem. The true labels for these

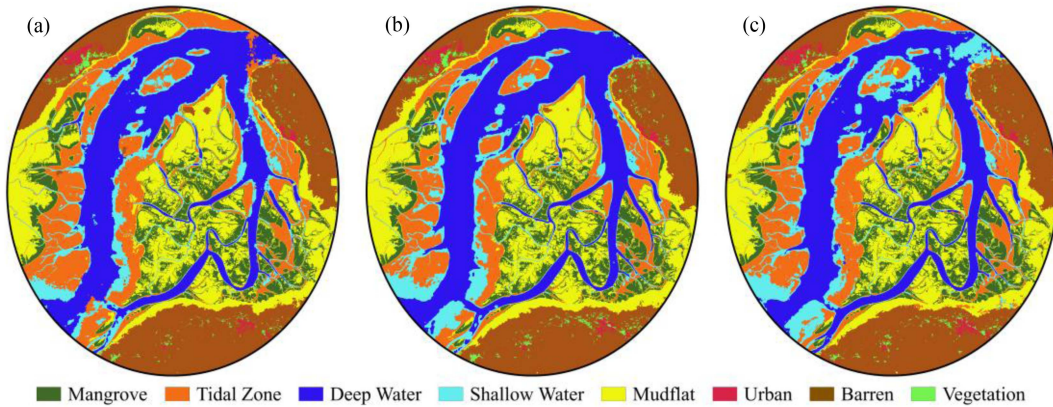


Fig. 11. Mangrove ecosystem maps for (a) 2017, (b) 2020, and (c) 2023 generated using the trained UNB7 model based on the satellite dataset and label mask of the baseline year.



Fig. 12. Zoomed-in areas illustrating mangrove expansion between 2017 and 2023, as depicted by the trained UNB7 model in the baseline year of 2020. Green, red, and blue colors represent mangroves in 2017, 2020, and 2023, respectively.

points were assigned using available high-resolution imagery, and the predicted labels were compared with the actual ones through the calculation of confusion matrices. Accordingly, the mangrove ecosystem maps for 2017 and 2023 achieved overall accuracies (OA) of 90.35% and 91.10%, respectively, demonstrating both high classification performance and promising temporal transferability of the UNB7 model. Further analysis revealed that the confusion patterns observed in the baseline map (Table I) were also present in the 2017 and 2023 results. Notably, misclassifications were more prominent among highly dynamic land cover types such as tidal flats, deep water, and shallow water. Despite these challenges, mangrove areas in both 2017 and 2023 were accurately delineated, with average PA and UA exceeding 95%, indicating the high applicability and reliability of the proposed approach for long-term monitoring.

Overall, both maps had minor changes in comparison to the map in the baseline year when considering the permanent land covers. It should be noted that although the satellite datasets were prepared similarly, the tidal situation between years was not identical, and considering classes with high transition characteristics could bias the change monitoring. Therefore, classes with high transitional variability, such as tidal zone, deep water, and shallow water, were excluded from the comparison. Expansion of mangrove extent was observed between the years, with expansion rates of 0.18% and 3.25% between 2017 and 2020 and 2020 and 2023, respectively (see Fig. 12). The expansions mostly occurred along the edges of land areas (on mudflats) and near denser mangrove patches, as similarly reported by [73]. Notably, the sparse mangrove patches with open canopies that

were not detected in earlier years were distinguished later as they grew and the mangrove density increased.

IV. DISCUSSION

Detailed mangrove ecosystem maps are valuable resources that contribute to achieving at least three of the United Nations Sustainable Development Goals (SDGs) of Climate Action (SDG13), Life in Water (SDG14), and Life on Land (SDG15; [74]). The former aims to mitigate the impact of climate change and is associated with the high potential of carbon sequestration in the mangrove ecosystems (i.e., mangroves and mudflats). The latter two focus on promoting sustainable use of terrestrial and aquatic ecosystems, highlighting the preservation of ecosystem services provided by mangrove ecosystems, such as habitat for biodiversity, coastal protection, and water filtration. Only a few recent studies have incorporated deep learning models and remote sensing datasets for detailed mangrove ecosystem mapping, with most of them focusing on multispectral images [31], [32], [33]. Accordingly, further investigations are needed to assess the applicability of SAR and deep learning models for detailed mangrove ecosystem mapping, especially in cases where these ecosystems are mostly located in cloud-prone tropical and subtropical areas. Such maps are instrumental in helping managers successfully enact legislation to support conservation practices and the rehabilitation of mangroves [13], [75]. Moreover, these maps can be used in habitat restoration initiatives through identifying and prioritizing regions for replantation efforts. For instance, identifying mudflat areas where aerial roots

expand and mangroves grow can help allocate suitable locations for mangrove seedling plantations, contributing to blue carbon development in the context of sustainability. Furthermore, the policymakers benefit from these maps in developing strategies to enhance resilience against coastal erosion, preventing carbon emissions from degraded ecosystems.

Visual interpretations, statistical assessments, and comparisons with high-resolution satellite images confirmed the promising performance of the proposed workflow for mangrove ecosystem mapping. It is worth noting that a patch size of 64×64 was used for the analysis due to achieving higher accuracy, while other tested patch sizes of 32×32 , 128×128 , 256×256 , and 512×512 yielded lower accuracies for this task. This is due to varying levels of contextual information and the inclusion of heterogeneous pixel groups associated with different patch sizes, as explored in previous literature [76], [77]. The adopted ensemble strategy, using the eight-fold predictions, resulted in a smoother map compared to regular patch concatenation. However, the eight-fold prediction and weighted concatenation entailed higher computational time, though it has a negligible impact when no real-time predictions are required. Notably, individual assessment of each fold of the eight-folded satellite dataset revealed the practicality of considering all viewing angles to reach higher accuracies. In particular, the generated mangrove ecosystem maps based on various one-folded datasets obtained an average OA of 94.63%. Moreover, the UNB7 model within the developed framework showed great capability in alleviating salt-and-pepper noises present in the label mask, which arose from pixel-based classification. This highlights the applicability of weak supervision for generating accurate mangrove ecosystem maps using deep learning models and nonideal label masks. The weak supervision eliminated the need for collecting highly accurate reference samples, which is laborious and cost-intensive [78], while also providing numerous reference samples to support developing deep learning models, such as UNB7. Furthermore, relying only on SAR satellite datasets allows frequent monitoring of mangrove ecosystems, even in cloud-prone areas with diverse climate conditions in which multispectral images are less practical. Here, an annual satellite dataset was prepared for mangrove ecosystem mapping, though it is believed that the model has the potential to be employed in shorter periods. For instance, the model could be employed for bimonthly mangrove ecosystem mapping and monitoring if suitable SAR images are captured from the study area with diverse tidal situations. Moreover, the developed model has the potential to generate mangrove extent maps on a biweekly basis with minimal impact from the tidal situation in regions where mangroves are partially submerged.

Visual comparison of the extracted mangrove extent with the third version of GMW (GMW V3; [72], [79]) revealed the superior performance of the UNB7 model in mangrove delineation. Despite the varying spatial resolution of input SAR images in the two studies, the GMW V3 maps were generated using the XGBoost classifier, and a deep learning model (U-Net + EfficientNet-B7) was used in the current study. The results manifested the superiority of the deep learning model over conventional machine learning algorithms, such as XGBoost

and RF (as supported by this study). The lower accuracy of GMW V3 emphasizes the necessity of conducting regional and local mangrove studies since the global results may include high uncertainties.

The temporal transferability of the UNB7 model was evaluated by generating mangrove ecosystem maps for 2017 and 2023 using the trained model in the baseline year (i.e., 2020), which proved to be acceptable based on visual interpretations and statistical assessments. Such an approach can eliminate the need for frequent reference sample collection as a resource-intensive part of the regular mapping and updating procedure. However, the predictive uncertainty analysis revealed that the average Shannon entropy increased from 0.1879 in 2020 to 0.2417 and 0.2727 in 2017 and 2023, suggesting the lower confidence of the trained UNB7 model in assigning labels to each class. The model's performance in 2017 and 2023 could be further enhanced if a limited number of reference samples and sample patches were available for fine-tuning the model. Moreover, it is believed that the performance could improve in an active learning framework by fine-tuning the model with the predicted mangrove ecosystem maps of either 2017 or 2023 in the corresponding years [80]. In this case, the patches with lower uncertainty can be selected across the study area to fine-tune the model in an active learning framework.

Despite the promising performance of the proposed workflow, several limitations were identified. In particular, the model was not capable of delineating very sparse and scattered mangroves with a dominance effect of mudflats within the 10-m spatial resolution of Sentinel-1 [see Fig. 6(d)]. Moreover, since the study area had a limited portion of urban areas, relatively fewer samples (patches) of this class were fed to the model during the training phase, which could be a reason for the lower accuracy for the urban classification. This issue could potentially be tackled either through adding additional sample patches with urban samples using data augmentation or Generative Adversarial Networks to expand the training patches of a specific class [81]. In this study, only SAR features extracted using EfficientNet-B7 were used for the semantic segmentation task; however, in future studies, adding hand-crafted diverse SAR features according to the sensor characteristics could improve the results [82], [83].

V. CONCLUSION

Reliable and detailed mangrove ecosystem maps are necessary to support the implementation of conservation and rehabilitation practices effectively. Comprehensive visual and statistical assessments confirmed the applicability of leveraging seasonal SAR images using the UNB7 (i.e., U-Net + EfficientNet-B7) deep learning model for detailed mangrove ecosystem mapping, where conventional machine learning algorithms (e.g., RF) fail to achieve high accuracy. The TTA ensemble strategy of weighted reconstruction of the final map from the 8-fold classified patches successfully eliminated artifacts and reduced misclassification errors at the patch borders, compared to regular patch concatenation. The achieved high OA of 95.56% demonstrated the practicability of using nonideal label masks in a weakly supervised framework alongside deep learning

models, highlighting their potential capability in alleviating salt-and-pepper noises. The comparison of the results with a recent global mangrove extent product (GMW V3) emphasized the necessity of conducting local and regional studies to generate more accurate maps. Moreover, the UNB7 model showed a potential temporal transferability by generating accurate mangrove ecosystem maps for 2017 (past) and 2023 (future), using the trained model in the baseline year of 2020, although a rise in uncertainty values was observed. This approach could reduce the need for frequent reference sample collection, simplifying map updates and providing ongoing insights into the study area's condition. The results of this study can support achieving SDG13, SDG14, and SDG15 by promoting biodiversity and ecosystem services monitoring through reliable mapping and temporal transferability.

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